

Isocolon, parison, and comparable-set construal: A corpus test of formal balance

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Abstract

Rhetorical theory often treats balanced form as more than ornament. A balanced pair of cola can present two units as candidates for comparison. A reader-side causal claim would need experimental evidence, but a production-side textual claim can be tested in corpus data. This paper tests whether adjacent discourse-relation pairs in the Georgetown University Multilayer corpus with enhanced Rhetorical Structure Theory annotations (GUM/eRST) show more **FORMAL BALANCE** when they construe two units as comparable co-members of a set. The analysis separates **ISOCOLON**, measured as length balance, from **PARISON**, measured as syntactic parallelism, and **LEXICAL ECHO**. In 17,456 adjacent relation pairs from 299 documents across 24 genres, a broad target family of comparison and co-ordination relations shows a positive adjusted increase in the composite formal-balance score. The largest broad target-family effects come from lists and disjunctions, where co-ordination is a strong grammatical null; the cleaner non-enumerative signal comes from adversative contrast.

The composite increase comes mostly from the parison subscore. Mean-only length controls also show isocolonic length balance, but they contribute little to the main composite estimate. A matched co-ordination check shrinks the list/disjunction effect; an adversative-only check leaves contrast positive. Antithesis alone sits near zero, and concession is negative. The result supports a restrained textual claim: formal balance belongs to the profile of some comparable-set construals and varies across relations that could be called contrastive.

1 INTRODUCTION

Rhetorical theory has long treated figures as patterned forms with argumentative force. In that tradition, **ISOCOLON** joins two kinds of evidence that are easy to confuse: formal balance between adjacent units and a reader's sense that the units invite comparison. The motivating claim is simple. When writers place two clauses or phrases in balanced form, they may present the comparison between them as more explicit, pointed, or forceful.

This paper asks how much of that claim can be tested without a reader experiment. The answer depends on the explanatory level. At the reader-response level, the claim says that isocolon changes

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what readers notice or how forcefully they construe a relation. Corpus evidence alone can't show that. At the textual level, the claim turns distributional: writers may preferentially use balanced form where they construe two units as comparable. That claim is testable.

The corpus test uses a narrow hypothesis. When a relation places two adjacent units in a shared comparison frame, the pair should show more formal balance than pairs expressing cause, condition, elaboration, sequence, or other non-comparative relations. The study uses the Georgetown University Multilayer (GUM) corpus and enhanced Rhetorical Structure Theory (eRST) annotations because they provide genre-diverse texts, discourse-relation labels, and signal annotation (Zeldes, 2017; Zeldes et al., 2025).

The empirical object is therefore a constructed composite formal-balance score. For every adjacent pair, that composite asks how much the two units resemble each other in length, surface syntax, and lexical material. It's graded, not binary: high composite values mark strong formal resemblance, not confirmed rhetorical figures.

The analysis uses the eRST relation export: mostly GUM, plus a smaller GENTLE challenge set included in the same export. Access to the Rhetorical Structure Theory (RST) Signalling Corpus is still pending, so that corpus isn't part of this analysis. Once available, it can support later triangulation against a broader inventory of coherence-relation signals.

The pattern refines the rhetorical claim. Broad target-family relations are more formally balanced than matched non-target relations, but the largest broad target-family effects occur exactly where a grammar-driven co-ordination account is strongest: lists and disjunctions. Ordinary adversative contrast is smaller but cleaner, because it survives inside an adversative-only comparison. Antithesis alone is near zero, and concession is negative. The pattern suggests that formal balance, especially parison-like syntactic parallelism, tracks some acts of comparative construal better than it tracks the inherited relation name ANTITHESIS.

2 RHETORICAL FIGURES AS FORM-FUNCTION PAIRINGS

The analysis starts from Fahnestock's account of rhetorical figures as form-function pairings. In her work on scientific argument and rhetorical style, figures are recurrent formal arrangements that can epitomize lines of reasoning (Fahnestock, 1999, 2011). That view gives the corpus prediction. If a figure serves an argumentative operation, then the figure should be distributed unevenly across the discourse contexts where that operation is being performed.

Fahnestock's discussion of the figures of argument speaks directly to this project. ISOCOLON concerns approximate equality in the length of adjacent cola, while PARISON concerns similarity of grammatical structure.

The two often co-occur, but the properties differ. A pair can be close in length without being syntactically parallel, and it can be syntactically parallel without being closely balanced in length. The classical tradition links both to comparison and contrast because similar form can make two units cohere as an argumentative set (Fahnestock, 2011).¹

A figure can help make a relation salient without being obligatory whenever the relation is present. For the corpus test, uneven distribution is expected, and frequency by itself gives no direct measure of the rhetorical value of the figure.

¹I approach these terms as a corpus linguist with recent rhetorical reading. That outsider position matters: the Greek terms motivate hypotheses, but the corpus tests the mappings. Less formally, this study stays suspicious of its own Greek.

The same point motivates computational work on rhetorical figures. Harris and Di Marco (2017) argue that figural form is often iconically related to discourse function, so surface patterning can provide evidence about pragmatic and argumentative actions. Harris et al. (2018) develop an annotation scheme for figures in part because figure identification requires attention to both formal regularity and rhetorical role. Lawrence et al. (2017) make the parallel point for argument mining: rhetorical figures can identify regions where argumentative work is being done, but only if formal detection is connected to the content and relation being advanced.

Together, these studies motivate the working assumption. The analysis pairs surface balance with a discourse relation that makes two units available for comparison. The operational question asks whether formal balance, separated into isocolonic length balance, parison-like syntactic parallelism, and lexical echo, is higher in the relations where rhetorical theory predicts that balance has value.

3 FROM CANDIDATE FORM TO RHETORICAL FUNCTION

The primary methodological challenge concerns over-attribution. Repetition and parallelism are common in ordinary text, and only some instances have the rhetorical force that figure theory cares about. Work on automatic chiasmus detection makes the problem vivid. Dubremetz and Nivre (2018) show that trivial repetition patterns are easy to find, but repetitions with rhetorical effect are much harder to separate from accidental or irrelevant patterns. They treat figure detection as a ranking problem instead of a clean binary classification problem. The problem is not finding parallelism. The problem is persuading it to mean something.

That lesson carries over to isocolon and parison. Exact repetition, list formatting, dialogue formulas, quoted slogans, and segmentation artifacts can all look formally balanced. Some are rhetorically relevant, but many aren't instances of the target phenomenon. Recent survey work makes the broader point: rhetorical figure detection remains limited by small datasets, inconsistent definitions, and over-reliance on rule-based surface cues (Kühn & Mitrović, 2024; Kühn et al., 2024). A credible corpus test should use surface measures cautiously, validate them against human judgment, and avoid treating high balance as equivalent to a gold figure label.

The analysis below follows that strategy. The composite formal-balance score is a continuous outcome in the distributional analysis and a ranking score in validation. It ranks adjacent relation pairs by how much measurable formal balance they display. The qualitative audit then asks whether top-ranked examples are plausible formal-balance candidates after excluding exact repetition, dialogue formulas, and other false-positive types. This more modest target than automatic figure recognition is the right target for a first rhetorical-function test.

4 CORPUS AND MEASUREMENT

4.1 THE CORPUS AND WHAT COUNTS AS A PAIR

The corpus comes from the English Georgetown University Multilayer corpus with enhanced Rhetorical Structure Theory annotations (GUM/eRST). GUM provides genre-diverse, richly annotated English texts (Zeldes, 2017). eRST extends RST-style discourse annotation with graph structures and explicit signal annotation across spoken and written genres (Zeldes et al., 2025). The DISRPT shared-task export used here also includes a smaller GENTLE challenge set. The project needs discourse relations as well as adjacent sentences. It also needs enough genre variation to separate a rhetorical effect from a register-specific habit.

The analysis uses adjacent discourse-relation pairs. The dataset contains 17,456 such pairs from 299 documents across 24 genres. These discourse units serve as candidate cola-like spans, with no claim that they’re classical cola in a strict prosodic sense.

A relation enters the analysis if eRST marks it as explicit or implicit, both spans have token intervals, and the two intervals are adjacent in document order. Non-adjacent linked spans are excluded. Concurrent graph relations are kept as pairwise relation rows when their spans meet this adjacency criterion.

The DISRPT shared-task relation export used here omits the full eRST graph-node inventory from the derived table. The diagnostics in Table 1 are interval-level checks instead of a complete graph-theoretic audit. The extracted set contains 16,278 rows with collection label GUM and 1,178 with collection label *GENTLE*, the out-of-domain challenge set; the 24 genres are the document-genre labels in the extracted files.

Extraction diagnostic	Count
Valid explicit/implicit relation rows	34,976
Adjacent relation rows used in analysis	17,456
Non-adjacent relation rows excluded	17,520
gapped rows excluded	15,349
interleaved rows excluded	2,171
Adjacent rows with discontinuous token intervals	1,963
Exact duplicate adjacent span pairs	0
Adjacent explicit rows	4,943
Adjacent implicit rows	12,513

Table 1: Extraction diagnostics for the adjacent-pair analysis. Counts come from token intervals in the DISRPT shared-task relation export. Exact duplicate span pairs are counted by document, unit-1 tokens, and unit-2 tokens.

4.2 THREE MEASURES OF FORMAL BALANCE

For each pair, I computed three surface measures. First, ISOCOLON measures length balance over token, character, and estimated syllable length. Second, PARISON measures similarity in dependency and lexical-category profiles. Third, LEXICAL ECHO measures repeated or closely corresponding lexical material. The composite formal-balance score combines the three subscores as a graded measure of formal balance. That composite gives evidence about formal balance, not a figure label for the pair.

The subscore calculations stay deliberately simple. For any two units with positive length, a component value is $1 - |n_1 - n_2| / \max(n_1, n_2)$. The value is 1 when the two units match exactly on that dimension and falls toward 0 as they diverge. Two units of 8 and 10 words, for instance, score $1 - |8 - 10| / 10 = 0.8$ on word length. Two units of equal length score 1. The isocolon subscore averages this quantity over word tokens, characters, and estimated syllables.

The parison subscore compares the two units’ grammatical make-up rather than their wording. It averages edit similarity over three sequences, each read in token order: the coarse Universal Dependencies lexical-category tags (*UPOS*), the finer language-specific ones (*XPOS*), and the dependency relations, which name the grammatical function (subject, object, modifier, and so on) that ties each word to the word it depends on.

Edit similarity counts how few small changes, inserting, deleting, or swapping one item, turn one such sequence into the other, rescaled to lie between 0 and 1 (Levenshtein, 1966). Two units

that run through the same categories in the same order therefore score near 1. Because the sequences are read flat, left to right, the measure registers shared category order but not the branching shape of the sentence; it's a surface profile, not a comparison of full parse trees.

The lexical-echo subscore applies the same comparison to the words themselves, in their lemma form, the dictionary form that counts *run*, *runs*, and *running* as one word. It looks only at content words, the meaning-bearing nouns, verbs, adjectives, and adverbs, setting function words such as *the* or *of* aside. Two quantities are averaged: edit similarity over the content lemmas in order, and Jaccard overlap, the share of content lemmas the two units have in common regardless of order (Jaccard, 1901).

The GUM/eRST files use the Conference on Natural Language Learning format for Universal Dependencies (CoNLL-U), which records tokenization, lemmas, lexical-category tags, and dependency labels. Punctuation tokens are excluded from the word, syntax, and lexical sequences, while colon and semicolon position are kept as separate controls. Syllable counts are approximate vowel-group counts.

The composite formal-balance score equals

$$0.40 \times \text{isocolon} + 0.45 \times \text{parison} + 0.15 \times \text{lexical echo}.$$

These weights define only that composite. The subscores are also reported separately, so the analysis can distinguish isocolonic length balance from parison-like syntactic parallelism and lexical echo.

4.3 DEFINING THE TARGET RELATIONS

At this point, the composite formal-balance score is only a property of each adjacent pair. It says how closely the two units match in length, surface syntax, and lexical material. It doesn't say that the pair belongs to a target relation, and it doesn't by itself identify a rhetorical figure. The target definitions below do separate work: they say which eRST relation labels should, by hypothesis, tend to have higher composite formal-balance scores.

The next step is to sort the eRST relation labels into hypothesis groups. Table 2 gives the mapping. These groups are analysis groups, not new annotations: they say which existing eRST labels count as targets for this test. The broad target family includes *adversative-antithesis*, *adversative-contrast*, *joint-disjunction*, and *joint-list*. This family tests whether formal balance rises in relations that place units in a comparable or co-ordinated frame. The narrow adversative family includes only *adversative-antithesis* and *adversative-contrast*. Individual-relation analyses separate the broad target-family effect into its individual labels.

Analysis role	Labels, rationale, and caveat
Broad target family	<i>adversative-antithesis</i> ; <i>adversative-contrast</i> ; <i>joint-disjunction</i> ; <i>joint-list</i> . Contrast, alternatives, and co-enumeration place units into a shared frame. The joint labels are co-ordination-rich and also the strongest grammar-driven null.
Narrow adversative family	<i>adversative-antithesis</i> ; <i>adversative-contrast</i> . Tests whether the broad result survives without list and disjunction labels.
Cleaner non-enumerative test	<i>adversative-contrast</i> . Tests contrast against other adversative relations, where list syntax is not the main explanation.
Contrastive control	<i>adversative-concession</i> . Adversative but often asymmetric; predicted to be weaker or negative.
Surface positive control	<i>restatement-repetition</i> . Expected to score high by form alone; used as a false-positive diagnostic, separate from the target hypothesis.

Table 2: Mapping from eRST relation labels to analysis roles. The table defines the operational grouping. Independent annotation of comparable-set construal remains future work.

4.4 ESTIMATING THE EFFECTS

The primary comparisons ask a plain difference question: after removing obvious confounds, do target relations receive higher composite formal-balance scores than other relations? The controls remove differences that can make the comparison too easy: document, explicit signalling, sentence adjacency, punctuation, mean length, and maximum length. Document controls mean that, where possible, target and non-target pairs are compared inside the same text.

For readability, I report effects as score-point differences on a 0–100 version of the composite formal-balance scale, where a pair sits near 0 when its two units share little measurable form, with different lengths, structures, and words, and near 100 when they match closely on length, syntax, and wording.

Figure 1 shows the empirical range behind that scale and splits it by broad target status. The 0–100 scale is a readability transformation, not a claim that the observed data fill the whole interval evenly. In the 17,456 adjacent pairs, the median composite value is 32.8. The middle half runs from 22.3 to 41.9, and 95% of pairs fall below 55.9. A few exact or near-exact matches reach the upper tail, but most of the empirical scale is the lower and middle part of the formal-balance range. The broad target family is visibly shifted right, but the two distributions overlap heavily.

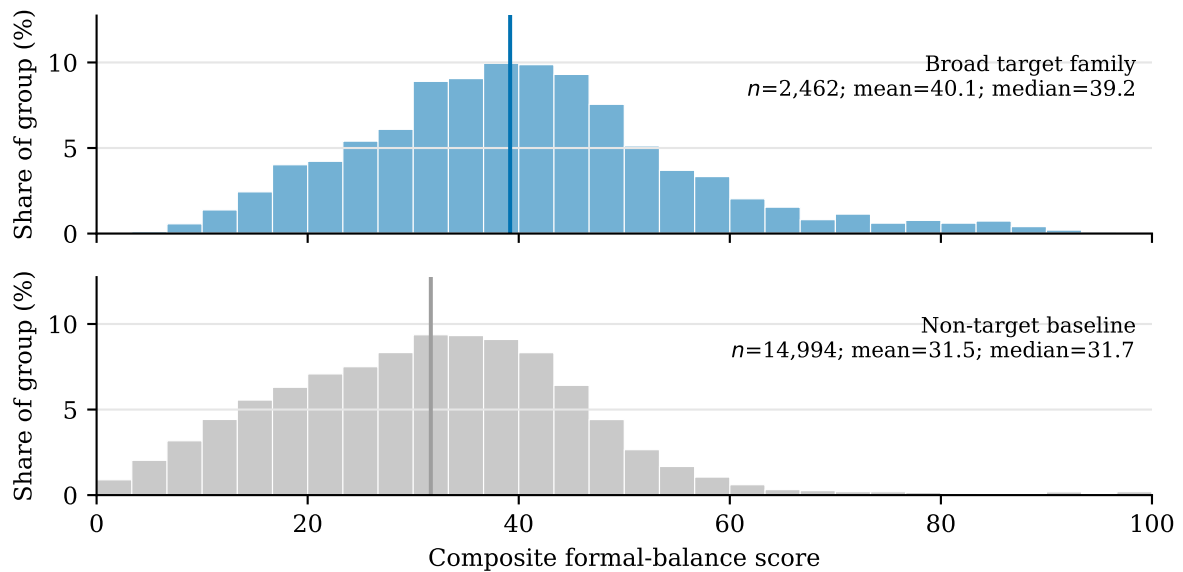


Figure 1: Distribution of the composite formal-balance score after rescaling to 0–100, split by broad target status. Bars show percentages within each group, not raw counts, so the larger non-target baseline doesn’t dominate the display. Vertical lines mark group medians. The plot is descriptive; the 4.1-point effect reported below is adjusted for document, length, punctuation, and other controls.

For scale, the most extreme comparison isn’t a target relation at all. It is simple restatement or repetition: after controls, repeated pairs sit 19.0 points above other pairs. One such pair, *Get your feet off of there.* followed by *Get your feet off of there.*, receives a composite formal-balance score of 100.0. That’s 68.3 points above the non-target median of 31.7. Table 3 gives additional anchors for

the raw per-pair score scale, with one target example and one non-target example near each anchor.

Score	Status	Gap	Unit 1	Unit 2
95.2	Target	+63.5	<i>You can use this on twist outs.</i>	<i>You can use this on braid outs.</i>
100.0	Non-target	+68.3	<i>Get your feet off of there.</i>	<i>Get your feet off of there.</i>
59.9	Target	+28.2	<i>I worked with some great musicians</i>	<i>and performed around the world.</i>
60.0	Non-target	+28.3	<i>I got to go out on the road.</i>	<i>I want to get my chops back up.</i>
50.0	Target	+18.3	<i>the house festooned with flowers</i>	<i>and stocked with coconut liquor.</i>
50.0	Non-target	+18.3	<i>it forces you to take action,</i>	<i>as you can't afford to waste any time.</i>
40.0	Target	+8.3	<i>maybe I should vlog today</i>	<i>and just see how my day turns around</i>
40.0	Non-target	+8.3	<i>Steven measured and mixed the plaster of Paris</i>	<i>and prepared for the cast</i>
30.0	Target	−1.7	<i>Shorts and sleeveless shirts are highly inappropriate,</i>	<i>as are short skirts.</i>
30.0	Non-target	−1.7	<i>For your joke to be funny,</i>	<i>it has to surprise the audience in some way,</i>
19.6	Target	−12.1	<i>We have no choice</i>	<i>but to return to direct negotiations between the two parties.</i>
20.0	Non-target	−11.7	<i>Yeab no.</i>	<i>Not a good look.</i>

Table 3: Example raw composite formal-balance scores. Gap is measured in score points against the non-target median of 31.7. TARGET means the pair belongs to the broad target family; NON-TARGET means it doesn't. These examples calibrate the raw scale; they aren't adjusted effects.

The broad target-family gap is 4.1 points, about a fifth of that repetition gap. The empirical standard deviation of the composite formal-balance score is 14.8 points, so the broad target-family gap is about 0.28 standard deviations. The effect is therefore modest.

What earns the 4.1-point gap a place in the paper isn't its size but its steadiness. The interval reported in Results stays well clear of zero.

I estimate these differences with ordinary least squares regression. In plain terms, it reports the average gap in balance between target and non-target pairs that are otherwise alike, holding the controls (document, length, punctuation, sentence adjacency, and explicitness) constant, so the gap isn't just a by-product of those background differences.

For the composite formal-balance score, the controls matter because longer spans have more opportunities for structural mismatch. The length subscore needs gentler handling. A pair's token-length balance is almost fixed once its mean and maximum span length are known. It equals $2\mu/m - 1$, where μ is mean span length and m is maximum span length. Holding both lengths constant at once would therefore subtract away most of the quantity being measured. The isocolon subscore is reported with mean-length control and without maximum-length control.

The main controls are an intercept, explicit relation status, same-sentence adjacency, colon punctuation, semicolon punctuation, standardized log mean length, standardized log maximum length, and document fixed effects. The two length terms enter logged and standardized: taking the logarithm keeps a few very long spans from dominating, and standardizing puts mean and maximum length on a common scale. Document-clustered intervals treat pairs from the same document as related observations rather than as wholly independent pieces of evidence (Cameron & Miller, 2015).

A genre-varying analysis asks whether the effect belongs to only one or two genres. It first estimates the same adjusted effect separately within each genre. Those single-genre figures are shaky wherever a genre contributes few pairs, so a second step (a normal empirical-Bayes model) pulls each one toward the overall average (Efron & Morris, 1973), moving the estimates built on little data the

most and the well-supported ones hardly at all. The same step reports how far the genres still spread around that shared average once the correction is made.

Robustness checks add mean-only length models, joint-only and adversative-only comparisons, GUM-only and continuous-span-only subsets, leave-one-genre influence checks, composite-weight sensitivity checks, and stratified permutation checks. In the permutation checks, the target labels are shuffled inside matched strata. The question is whether effects this large appear when relation labels are randomly assigned within comparable document, length, punctuation, and explicitness groups.

5 VALIDATION

The validation asks whether the composite formal-balance score ranks useful cases near the top. It doesn't ask whether the composite can label figures automatically. The eRST *syn-prl* signal labels mark cases where annotators identified syntactic parallelism as a discourse signal. Those labels give a small check on formal parallelism, not on rhetorical force.

I report the area under the receiver-operating characteristic curve (AUC) because it has a plain ranking interpretation that needs no statistical background (Hanley & McNeil, 1982). Imagine drawing two pairs at random: one that annotators marked with a *syn-prl* parallelism signal, and one they left unmarked. The AUC is the chance that the composite formal-balance score is higher for the marked pair than for the unmarked one.

A ranking that sorted the two no better than a coin toss would give an AUC of 0.5; a ranking that always put the marked pair on top would give 1.0. So the AUC of 0.73 reported just below means that, in this head-to-head, the marked pair wins about 73 times in 100. The measure takes its name from a curve that plots recall against the false-positive rate at every cutoff, but the simple ranking reading is all that's needed here.

On that scale, the composite formal-balance score has an AUC of 0.73, with an approximate Hanley–McNeil interval from 0.66 to 0.80. The parison subscore is slightly higher at 0.75 [0.69, 0.82], the lexical-echo subscore is 0.72 [0.65, 0.79], and the isocolon subscore is 0.65 [0.58, 0.72] (Table 4). The interval gives a rough uncertainty range for the AUC, not a new effect size.

A second set of numbers asks a more practical question. If a researcher read only the top-ranked pairs, hunting for genuine parallelism to audit, how rewarding would that slice be? Take the top 5% of pairs by the composite formal-balance score: everything at or above the 95th percentile.

Three quantities describe that slice. PRECISION is the share of the slice that actually carries a *syn-prl* label: here about 3 pairs in every 100. RECALL is the share of all the labelled pairs in the corpus that the slice manages to gather up: here 37%, so reading just the top 5% already recovers more than a third of the parallelism signals. LIFT compares that hit rate to how common the label is across the corpus as a whole.

Because *syn-prl* labels are rare, only 73 among 17,456 pairs, well under one in a hundred, a 3% hit rate in the top slice works out to a lift of 7.4. The slice is roughly seven times richer in labelled pairs than a random handful would be. Tightening the cut to the top 2.5% (the 97.5th percentile) raises precision to 5.5% and lift to 13.1, while recall settles at 33%.

Those numbers look poor for automatic labelling but adequate for ranking rare candidates for audit. The validation also stays narrow: the *syn-prl* labels validate a surface parallelism signal. In this implementation, a pair is positive if either unit contains a token-level *syn-prl* signal in the eRST export, so the validation uses span-level signal presence as a proxy for relation-level syntactic parallelism. It

doesn't validate rhetorical function itself.

Measure	AUC	Gold mean	Non-gold mean
Composite formal balance	0.73 [0.66, 0.80]	0.50	0.33
Isocolon	0.65 [0.58, 0.72]	0.68	0.56
Parison	0.75 [0.69, 0.82]	0.44	0.21
Lexical echo	0.72 [0.65, 0.79]	0.20	0.06

Table 4: Validation against the 73 eRST *syn-prl* signal labels in 17,456 adjacent relation pairs. AUC gives the chance that a random labelled pair receives a higher value than a random unlabelled pair. The gold-mean and non-gold-mean columns are the average values for pairs that do and don't carry the label, on the same 0–1 scale as the composite formal-balance score and the three subscores. Brackets give approximate Hanley–McNeil 95% intervals for AUC.

The qualitative audit supplies a pilot rhetorical-function check. I coded 80 candidate pairs in a single-author, non-blind audit; row strata and composite formal-balance values were visible. Automated reviews only flagged borderline cases after coding. The audit isn't interrater-reliability evidence. It identifies false-positive classes and illustrative examples.

The pattern is the point: top-ranked broad target-family candidates usually survived the audit, while repetition controls didn't. Exact repetition gives strong surface balance, so it functions as a control rather than a target.

The audit also clarified the boundary of the category. Bare identity, apposition, dialogue repair, and exact repetition can be rhetorically relevant apart from isocolon. Conversely, weak length balance can still matter when the two units share a frame strongly enough to invite slot substitution. The coding decisions treat isocolon as a component of a broader rhetorical-function judgment, with false-positive classes instead of a mechanically recoverable surface pattern.

Sample stratum	<i>n</i>	Yes	No	Unc.	Main issue
High broad target family	16	14	2	0	Slot substitution
Mid broad target family	12	7	4	1	Weak frames
Narrow adversative	14	6	8	0	Label mismatch
High non-target/artifact	19	4	12	3	Dialogue/apposition
Concession candidates	5	1	4	0	Asymmetry
Repetition controls	9	0	9	0	Exact identity
Baseline non-target	5	1	4	0	Weak or no frame

Table 5: Human qualitative audit summary for 80 candidate pairs.

Type	Unit 1	Unit 2	Judgment
Slot substitution	<i>You can use this on twist outs.</i>	<i>You can use this on braid outs.</i>	Yes
Adversative contrast	<i>so that the broad sides of the cuttings face east and west</i>	<i>while the slim sides face north and south.</i>	Yes
Repetition	<i>Get your feet off of there.</i>	<i>Get your feet off of there.</i>	No
Label mismatch	<i>We are not setting out to turn them into programmers or data crunchers,</i>	<i>but want to boost their knowledge level</i>	No

Table 6: Illustrative audit examples. The final row is an *adversative-antithesis* relation label. It lacks the balanced opposed cola of classical antithesis.

6 RESULTS

The broad target family is more formally balanced than the non-target baseline. On a 0–100 version of the composite formal-balance score, target pairs are 4.1 points higher after controls. The empirical standard deviation of that score is 14.8 points, so the adjusted gap is about 0.28 standard deviations. The document-clustered interval runs from 3.6 to 4.7 points. The shift is modest but stable.

The subscores show what kind of balance carries the effect. The parison subscore difference is 7.7 points, compared with 2.4 points for the lexical-echo subscore. With mean-length control only, the broad isocolon subscore difference is 6.6 points. Table 7 reports the composite formal-balance score and the three subscores separately.

Two columns in that table both describe length balance, and the gap between them is worth a word. The LENGTH column applies the full control set, including both mean and maximum span length. The ISOCOLON column reports the same length-balance outcome but drops the maximum-length control, for the reason given in the measurement section: with both length controls in place, almost none of the length-balance subscore is left to estimate. So the LENGTH column is a deflated, doubly conditioned figure (0.8 points for the broad target family), while the ISOCOLON column shows the length-balance effect on its own terms (6.6 points).

Read the ISOCOLON column as the substantive length-balance result, and the LENGTH column as a record of how little survives once length is held fixed twice over.

The broad target-family result is an omnibus check. Its strongest labels are also the labels where co-ordination makes parallel form easiest, so the cleaner rhetorical test comes from more focused comparisons. The narrow adversative family, *adversative-antithesis* plus *adversative-contrast*, shows a smaller 1.3-point composite difference, with an interval from 0.5 to 2.1 points.

The individual labels explain the difference. *joint-disjunction* has an adjusted composite formal-balance effect of 4.8 points, *joint-list* has an adjusted effect of 4.4 points, and *adversative-contrast* has the smaller but more diagnostic effect of 2.4 points. *adversative-antithesis* is near zero at 0.4 points, with an interval that crosses zero. I read this as a mismatch between a discourse-relation label and the classical figure. It gives no evidence against antithesis as a rhetorical resource. *adversative-concession* is negative at -1.3 points.

The relation-level scan also identifies an important positive control. *restatement-repetition* has the strongest adjusted composite formal-balance effect, 19.0 points, with especially large lexical-echo and parison effects. That's exactly what the composite formal-balance score should find: exact identity is highly balanced, with lexical echo stronger than the parison effect. The human audit shows why it needs separate treatment. Repetition can signal clarification, identity, repair, emphasis, or other rhetorical work; exact repetition remains distinct from balanced comparison. The analysis treats re-statement/repetition as a control and diagnostic, separate from the target hypothesis.

Genre-varying estimates point in the same direction. After this pooling, the broad target family's mean effect is 3.9 points across 20 genres. The *joint-list* estimate is 4.0 points across 17 genres, and *adversative-contrast* is 2.9 points across the genres where there's enough evidence to estimate it. The narrow adversative-family estimate is 1.2 points.

The genre-by-genre effects scatter only narrowly around those averages: their spread, one standard deviation, is about 1.1 points for the broad target family, 1.3 for *joint-list*, 0.9 for the narrow adversative family, and 0.7 for *adversative-contrast*. Because that scatter is small next to the effects themselves, the broad target-family effect isn't the work of one or two genres but recurs across many.

Target	Target n	Other n	Composite	Length	Isocolon	Parison	Lexical
Broad target family	2,462	14,994	4.1	0.8	6.6	7.7	2.4
Narrow adversative family	605	16,851	1.3	-0.1	4.7	2.4	1.7
<i>joint-list</i>	1,690	15,766	4.4	1.0	6.5	8.2	2.0
<i>joint-disjunction</i>	167	17,289	4.8	0.9	3.6	8.6	4.0
<i>adversative-contrast</i>	262	17,194	2.4	-0.0	7.3	4.5	2.8
<i>adversative-antithesis</i>	343	17,113	0.4	-0.2	2.5	0.8	0.9
<i>adversative-concession</i>	679	16,777	-1.3	0.1	1.1	-2.7	-0.7
<i>restatement-repetition</i>	321	17,135	19.0	4.6	17.0	27.3	32.5

Table 7: Selected adjusted score-point differences on the composite formal-balance score and its sub-scores, using a 0–100 rescaling of the outcome. Target n gives the number of target rows; other n gives the number of rows outside that target. ISOCOLON is length balance, PARISON is syntactic parallelism, and LEXICAL ECHO is lemma-sequence similarity plus content-lemma overlap. Composite, length, parison, and lexical estimates use document fixed effects and controls for explicitness, sentence adjacency, punctuation, mean length, and maximum length. The ISOCOLON column uses the same length-balance outcome but omits maximum length because mean and maximum length nearly determine the length-balance subscore. For that reason, the composite formal-balance score isn’t the weighted sum of the displayed ISOCOLON, PARISON, and LEXICAL ECHO columns. The restatement row is a positive control, separate from the target hypothesis.

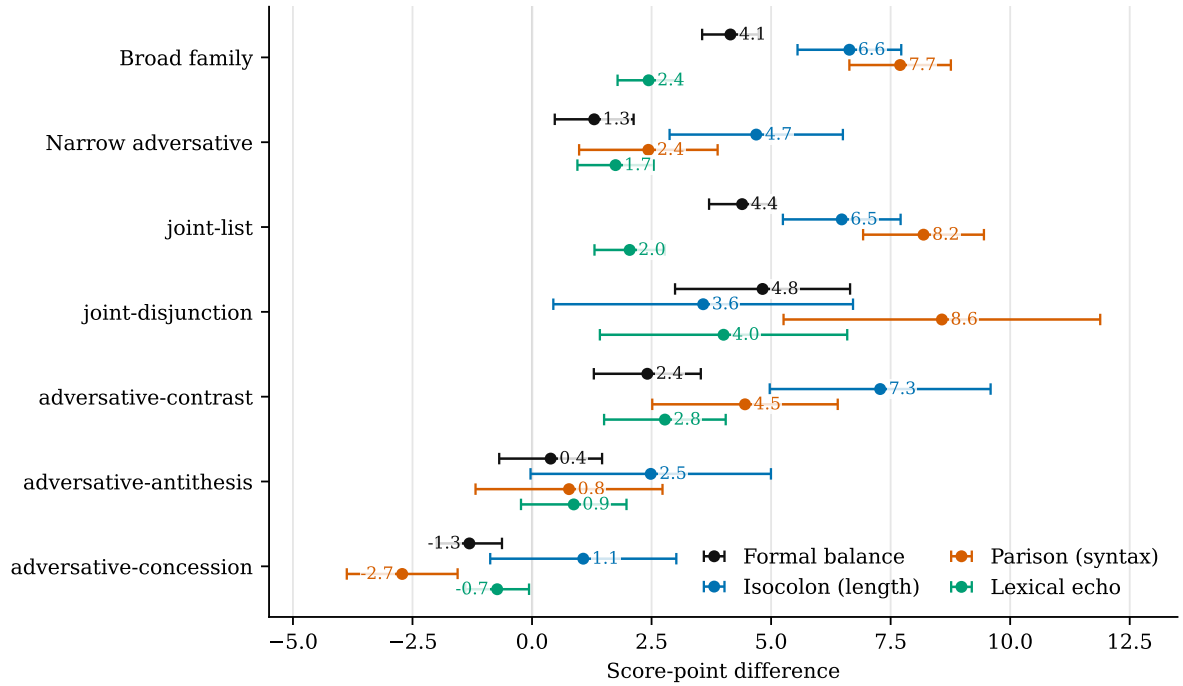


Figure 2: Adjusted score-point differences for the composite formal-balance score, the ISOCOLON subscore, the PARISON subscore, and the LEXICAL ECHO subscore. Points are estimates; horizontal intervals are document-clustered 95% intervals. Effects are shown as score-point differences on a 0–100 scale. The ISOCOLON estimates use mean-length controls and omit maximum length; the other estimates use the main control specification.

6.1 ROBUSTNESS CHECKS

Each check below removes one easy explanation for the main result and asks whether the effect survives. The reasoning is the same throughout. If balance only looked elevated because of some background difference between target and non-target pairs, then matching or stripping out that difference should make the effect shrink toward zero. Where the effect holds up instead, that particular explanation can be set aside. The checks move from the broadest (how the uncertainty is estimated) to the most demanding (a permutation test that scrambles the relation labels altogether).

The primary composite formal-balance results survive document-clustered uncertainty estimates. With document clustering, the broad target-family effect is 4.1 points, with an interval from 3.6 to 4.7. The *joint-list* effect is 4.4 [3.7, 5.1], *joint-disjunction* is 4.8 [3.0, 6.7], and *adversative-contrast* is 2.4 [1.3, 3.5]. *adversative-antithesis* remains near zero, 0.4 [−0.7, 1.5].

The isocolon subscore needs a different control specification. Mean and maximum span length nearly determine it. Using both controls therefore suppresses the quantity being estimated. With mean-length control but no maximum-length control, the broad target-family effect is 6.6 [5.6, 7.7], and *adversative-contrast* is 7.3 [5.0, 9.6]. The corresponding no-length-controls estimates are nearly identical: 6.7 points for the broad target family and 7.1 points for *adversative-contrast*. The isocolon-subscore claim is reported under mean-length control.

The obvious objection is the right one. Lists and disjunctions are easy places to be parallel: they line up coordinated members built to the same template, repeat frames, and often carry explicit co-ordination such as *and* or *or*. The joint-only comparison asks whether *joint-list* and *joint-disjunction* survive against other multinuclear relations: relations that join two units of equal standing rather than a main unit and a subordinate one. They do, but the effect shrinks to 1.7 points [0.6, 2.8]. The corresponding parison effect is 2.9 [1.0, 4.8], while the isocolon effect is only 0.3 [-0.7, 1.3].

The adversative-only comparison is cleaner. It compares *adversative-contrast* against other adversative relations, where list syntax isn't the main explanation. Contrast remains positive at 3.0 [1.4, 4.7] on the composite formal-balance score, again led by the parison subscore. The joint-only baseline consists of *joint-sequence* and *joint-other*; that baseline remains multinuclear and partly co-ordinative.

Two data-scope checks give similar results. In the GUM-only subset, the broad target-family effect is 4.0 [3.4, 4.6], the within-joint list/disjunction effect is 1.7 [0.6, 2.9], and the within-adversative contrast effect is 3.4 [1.6, 5.1].

Excluding the 1,963 adjacent rows with discontinuous token intervals leaves the broad target-family effect at 4.5 [3.8, 5.1], the within-joint list/disjunction effect at 1.8 [0.6, 3.0], and the within-adversative contrast effect at 3.3 [1.4, 5.2]. A leave-one-genre check also stays stable: the broad target-family estimate ranges from 4.0 to 4.3, the within-joint estimate from 1.5 to 2.0, and the within-adversative contrast estimate from 2.8 to 3.2.

The last and most demanding check asks whether the effect could be an accident of which relations happen to wear the target label. It uses a permutation test, and the logic is worth spelling out. Hold the texts, lengths, punctuation, and explicitness fixed. Then peel the relation labels off the pairs and deal them out again at random, keeping the same number of target labels inside each matched group of pairs.

Re-estimate the effect on this reshuffled data, where the labels now carry no real meaning. Repeat the whole procedure 500 times. The result is a picture of how large an effect can appear from label placement alone. If the real effect is no bigger than the shuffled ones, it could be an artifact of which pairs got the label; if it lands far beyond them, label placement alone can't account for it.

The shuffling respects strata defined by document, explicitness, same-sentence status, colon and semicolon punctuation, and quartile of mean word length, so labels are only ever swapped among genuinely comparable pairs. It doesn't hold local phrase type or root dependency category fixed, which is why it's a partial null rather than a complete one.

The observed broad target-family effect, the list/disjunction effect, and the adversative-contrast effect all land above the 97.5th percentile of their shuffled distributions (Table 9). Across 500 shuffles, no random relabelling produced an effect as large as the one actually observed. Because 500 shuffles pin down the tail only to about 0.002, I read this as a calibration check rather than a precise *p* value. The aim isn't formal significance testing but a demonstration that the effects sit well outside what label placement alone would produce.

A final check probes the composite weights themselves. The 0.40 / 0.45 / 0.15 split across length, syntax, and lexical echo was set by hand, so a fair worry is that the headline result depends on that one choice. To test this, the effect was re-estimated under 66 alternative weightings laid out on a grid that sweeps across the full range of nonnegative combinations of the three subscores. The broad target-family effect stays positive, with an interval clear of zero, in every one of the 66. For *adversative-contrast*, 65 of the 66 are positive and 59 have intervals clear of zero. The result doesn't hinge on the

particular weights chosen.

Check	Target/outcome	Estimate (points)	Clustered 95% interval
Main	Broad target-family composite	4.1	[3.6, 4.7]
Main	<i>joint-list</i> composite	4.4	[3.7, 5.1]
Main	<i>adversative-contrast</i> composite	2.4	[1.3, 3.5]
Mean-length only	Broad isocolon	6.6	[5.6, 7.7]
Mean-length only	<i>adversative-contrast</i> isocolon	7.3	[5.0, 9.6]
Joint-only	List/disjunction composite	1.7	[0.6, 2.8]
Joint-only	List/disjunction parison	2.9	[1.0, 4.8]
Adversative-only	Contrast composite	3.0	[1.4, 4.7]
Adversative-only	Contrast parison	5.9	[3.0, 8.8]
GUM-only	Broad target-family composite	4.0	[3.4, 4.6]
Continuous spans only	Broad target-family composite	4.5	[3.8, 5.1]

Table 8: Selected robustness checks, reported as score-point differences on a 0–100 scale. The joint-only rows compare *joint-list* and *joint-disjunction* against other *joint*-* relations. Mean-length-only rows control log mean length and omit log maximum length. The adversative-only rows compare *adversative-contrast* against other adversative relations. The continuous-spans-only row excludes adjacent rows where either discourse unit has a comma-separated token interval.

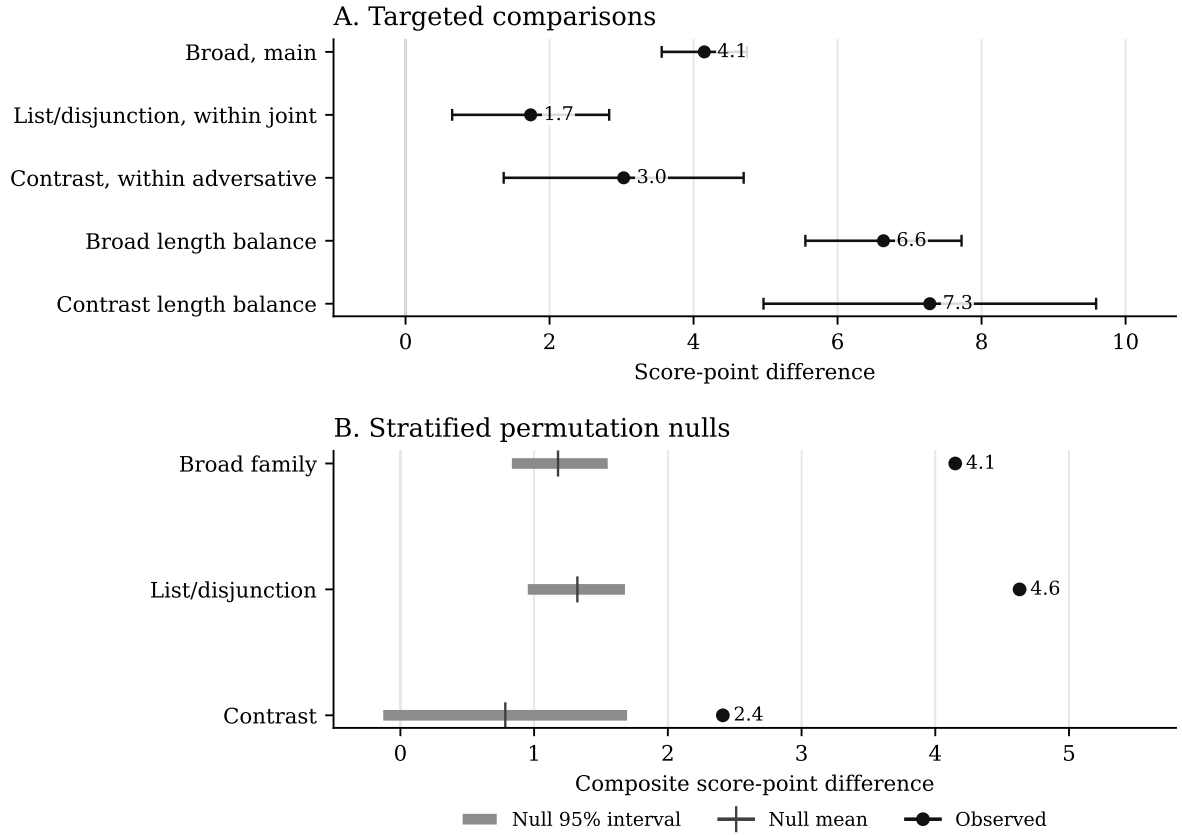


Figure 3: Targeted robustness checks and stratified permutation nulls, reported as score-point differences on a 0–100 scale. Panel A shows adjusted differences with document-clustered 95% intervals. Panel B compares observed composite effects with the null mean and central 95% interval from stratified target-label shuffles.

Target	Observed	Null mean	Null 95%	Strata	Rows	Target rows	<i>p</i>
Broad family	4.1	1.2	[0.8, 1.6]	1,240	8,277	2,174	0.002
List/disjunction	4.6	1.3	[1.0, 1.7]	1,033	7,047	1,659	0.002
<i>adversative-contrast</i>	2.4	0.8	[-0.1, 1.7]	201	1,472	220	0.002

Table 9: Stratified permutation nulls for the composite formal-balance effect, reported as score-point differences on a 0–100 scale. Target labels are shuffled within document, explicitness, same-sentence status, colon/semicolon status, and mean-length quartile strata. The strata, rows, and target-rows columns report the portions of the data that could actually move under the stratified shuffle. The *p* values are two-sided empirical values from 500 permutations, with the standard plus-one correction, and should be read at that resolution.

7 INTERPRETATION

The findings support a restricted version of the rhetorical-function claim. I treat balanced form as a selective way of presenting units as comparable. It's optional even when comparison is present. That view predicts an uneven distribution: balance should cluster where matched presentation has rhetorical value, while remaining absent in many ordinary cases. The corpus result fits that version, but unevenly. Lists and disjunctions supply the largest broad target-family effects; adversative contrast supplies the cleaner non-enumerative case.

COMPARABLE SET works here as a descriptive cover term for co-membership relations, including bare co-enumeration. The term allows list structure without treating every list as comparison for its own sake. The cleanest non-enumerative signal in the analysis comes from *adversative-contrast*.

The subscore results then refine the claim. The dominant effect looks parison-like more than isocolonic in the narrow length-balance sense. The evidence weakens for labels that look rhetorically relevant in the taxonomy but don't necessarily require balanced presentation in the text.

In particular, *adversative-antithesis* as an eRST relation label only partly overlaps with antithesis in the classical theory of figures. The label *adversative-antithesis* keeps the name while losing the balance.² The conservative conclusion leaves antithesis itself intact: the corpus label *adversative-antithesis* serves as a poor direct proxy for the classical figure.

That point remains a limitation as well as a finding. The 80-pair audit contains only eight rows with the *adversative-antithesis* label. These rows support the label-mismatch interpretation, but they're too few to separate label mismatch from a second possibility: genuine semantic antitheses may be expressed through opposition instead of lexical overlap or surface parallelism.

The subset test is defined but not yet reported here. The follow-up worksheet covers all 343 eRST *adversative-antithesis* rows and asks for three judgments: classical antithesis, semantic opposition, and parallel opposition. Once those fields are coded, the analysis compares yes and no rows on the composite formal-balance score and the three subscores. Until then, the near-zero *adversative-antithesis* result is evidence about the corpus label, not a settled test of classical antithesis itself.

The composite formal-balance score tracks a broader act of placing units into a shared frame. That frame can be contrastive, but it can also be co-ordinative, disjunctive, or list-like. In the latter cases, co-ordination itself favours matched daughters and weakens the rhetorical interpretation. Conversely, some adversative relations have little balance because their function is concession, qualification, or discourse management instead of matched presentation.

Corpus data record ordinary production, not an ideal of trained rhetorical practice. The evidence shows where balance appears in this corpus. It doesn't show where balanced form could improve a text. The language scope is also narrow. These are English corpus data, mostly from GUM, with English word order, English morphology, English punctuation, and English written and spoken segmentation. Nothing here licenses a claim about how isocolon or parison behave in Ancient Greek, Modern Japanese, or any other language whose prosody, morphology, discourse-unit segmentation, or rhetorical training traditions may differ.

The co-ordination caveat matters. Lists and disjunctions are places where parallel syntax is grammatically cheap, so the broad target-family effect can't be read as a pure rhetorical choice. The joint-only robustness check shrinks the effect, which is what that objection predicts. The effect remains

²The sentence is itself a small ISOCOLON and PARISON, and it asserts an ANTITHESIS. Either the author is demonstrating the apparatus or the apparatus is demonstrating the author.

positive, and *adversative-contrast* remains positive inside the adversative family and under a stratified permutation null.

The safer interpretation says that comparable-set relations attract formal balance beyond genre, document, length, and punctuation differences, while much of that balance follows the syntax of co-ordination. Relation labels don't guarantee classical figures, and classical figures needn't appear where relation labels promise them.

The pattern also clarifies the scope of the corpus study. The evidence stays textual and distributional. It concerns where balanced form occurs, not how readers experience those pairs. A high composite formal-balance score by itself still falls short of a rhetorical-figure label. The analysis shows that the composite formal-balance score is systematically higher in particular discourse-relation contexts. The human audit illustrates why many top-ranked cases are rhetorically relevant and why this remains pilot evidence, pending an independently validated annotation layer.

The RST Signalling Corpus isn't part of the analysis yet because access is still pending. Once available, that corpus can add a later triangulation point.

Work on discourse signalling has shown that coherence relations are often signalled by more than discourse markers, including lexical, syntactic, graphical, semantic, and genre-based cues. That point is developed in the original signalling annotation work and in later studies of RST signals and multiple signals (Das & Taboada, 2018a, 2018b, 2019; Taboada & Das, 2013). This literature gives the project a stronger comparative baseline. If formal balance behaves like a discourse signal, then it should be testable alongside other signal types rather than only as an isolated figure cue.

8 CONCLUSION

A corpus study can't prove that isocolon makes readers perceive comparison as more forceful, and it leaves rhetorical pedagogy untested. It can test a cleaner textual claim: writers use formal balance unevenly across discourse relations, and the relations most associated with comparable-set construals should show more of it. These results support that claim for the broad target family of comparison-like and co-ordinative relations, with the largest effects in lists and disjunctions and the cleaner non-enumerative effect in adversative contrast.

The analysis gives selective support to *ANTITHESIS* as an inherited figure label. The narrow adversative-family effect stays small, and the discourse-relation label *adversative-antithesis* sits close to zero. That finding sharpens the mapping between relation labels and classical figures. The strongest signal comes from parison-like parallelism. Isocolon narrowly understood as length balance contributes less to the main composite formal-balance effect. Balanced form remains a rhetorical resource, and the corpus evidence supports a claim about comparable-set construals and formal balance more securely than a claim about equal cola alone.

The robustness checks sharpen that conclusion. Length balance appears under mean-length conditioning without maximum length, while parison leads the main composite formal-balance model. The joint-only comparison also shows that a large share of the broad target-family effect comes from co-ordination itself. The stratified permutation check places the observed effects above document, length, punctuation, and explicitness-matched nulls. The result maps where figural form begins to do work.

DATA AND CODE AVAILABILITY

The analysis scripts, coding materials, and derived audit summaries are kept in the project repository: <https://github.com/BrettRey/isocolon-and-comparative-construal>. The raw GUM/eRST source files are excluded from the repository; the scripts expect a local corpus checkout under `data/raw/gum-erst`. The RST Signalling Corpus isn't included yet because access is still pending.

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